MIL-P-4641B
29 March 1960
Superseding
MIL-P-4641A(USAF)
28 June 1957

MILITARY SPECIFICATION

PLOTTING BOARD, WINDS ALOFT ML-514/TM

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers one type of plotting board for winds aloft data designated ML-514/TM.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification.

SPECIFICATIONS

Federal

L-V-351	Vinyl Chloride Polymer and Copolymer Rigid Sheets
NN-P-530	Plywood; Flat-Panel
PPP-T-60	Tape; Pressure Sensitive Adhesive, Waterproof - For Packaging and Sealing
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-591	Boxes, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Fiber

Military

JAN-P-100	Packaging and Packing for Overseas Shipment, General Specification
MIL-P-116	Preservation, Methods Of
MIL-B-138	Boxes, Wood, Fiberboard-Lined For Overseas Shipment (For Weight Contents Not Exceeding 500 Pounds)
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specifications For

FSC 6660

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MIL-N-7513	Nomenclature Assignment, Approval of Nameplate Identification and Descriptions; Aeronautical Support, Ground Electronic and Photographic Equipment, Contractors Method for Obtaining
MIL-B-10377	Box, Wood, Cleated, Veneer, Paper Overlaid
MIL-L-10547	Liners, Case, Waterproof
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-447	Definitions of Interchangeable Substitute and Replacement Items

PUBLICATION

Air Force-Navy Aeronautical Bulletin

No. 143	Specifications and Standards; Use of

(Copies of publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction.-This specification makes provision for preproduction testing.

3.2 Subassemblies.-The plotting board shall consist of:

Item No.	Quantity	Description	See Requirement
1	1 each	Base Board with permanently affixed laminated chart	3.4.1
2	1 each	Parallel attachment	3.4.1.2
3	1 each	Straightedge with scales	3.4.3
4	1 each	Frame	3.4.4

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3.3 Selection of specifications and standards.-Specifications and standards for necessary commodities and services not specified herein shall be selected according to ANA Bulletin No. 143, except as provided in 3.3.1 and 3.3.2.

3.3.1 Commercial parts.-Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, like screws, bolts, nuts, cotter pins, etc., having suitable properties, may be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawings.

3.3.2 Standard parts.-With the exception in 3.3.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.3.3 Metals.-Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion in atmospheric conditions likely to be met in storage or normal service.

3.4 Details of subassemblies

3.4.1 Baseboard

3.4.1.1 Plywood.-The baseboard shall be constructed of 5-ply, resin-bonded, 3/4-inch hardwood (birch or maple) plywood in accordance with NN-P-530. The board shall be 30 inches long and 30 inches wide, plus or minus 1/32 inch. The surfaces of the board shall be finished in such a manner as to permit compliance with 3.4.2.5.1 and 3.4.2.5.2.

3.4.1.1.1 Plastic backing.-A plastic backing with a minimum thickness of 0.010 inch shall be bonded to the underside of the baseboard. The backing shall be made of vinyl chloride acetate sheet material in accordance with L-V-351, shall cover the entire underside of the baseboard, and shall be opaque. Bonding shall be as specified in 3.4.2.5.1.

3.4.1.2 Parallel attachment.-The baseboard shall be equipped with a suitable attachment for maintaining the parallelism of the straightedge as the straightedge is moved up and down the board. This attachment shall be of the cord and pulley type and shall be located on the bottom surface of the plotting board. The material used for the cord shall not become frayed or

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introduce excessive slack into the operation of the straightedge with continuous use. When the plotting board is completely assembled, the cord shall not pass through the straightedge, but shall be attached to a pair of overhanging brackets (3.4.3.2) which are in turn secured to the straightedge. The parallel attachment shall permit the straightedge to be moved freely up and down the board, over the entire distance between the bottom and top metal edges.

3.4.1.3 Suction feet.-The baseboard shall be equipped with 4 rubber suction feet. These feet shall have an outside diameter at the base of not less than 1 1/4 inches, and shall be secured to the base of the parallel attachment by means of wood screws. The suction feet shall be located approximately 1 inch from each corner of the baseboard.

3.4.1.4 Folding legs.-The baseboard shall be equipped with two folding legs capable of lifting the back end of the board approximately 3 inches above the surface on which the board is placed. These legs shall support the board in the raised position evenly and firmly, and shall not interfere with the operation of the parallel attachment.

3.4.2 Laminated base chart

3.4.2.1 Construction.-The laminated chart shall consist of a paper chart laminated between two sheets of plastic (3.4.2.4). It shall be 30 inches long by 30 inches wide, plus or minus 1/32 inch, with a continuous border on all sides approximately 1/8 inch wide, formed by a plastic-to-plastic seal. The laminated chart shall be permanently bonded to the baseboard as specified in 3.4.2.5. The plastic sheets shall be as specified in 3.4.2.4.

3.4.2.2 Chart paper.-The paper used for the chart shall be 29 3/4 inches long by 29 3/4 inches wide, plus or minus 1/32 inch. High-quality, white enamel-coated book paper, 80 pound basis, entirely suitable for lamination shall be employed. The paper shall not be discolored or distorted by the lamination process.

3.4.2.3 Chart scales.-Six vertical scales and a horizontal base reference line shall be printed on the chart. Details of the scales and base line shall be as shown in figure 1 and herein.

3.4.2.3.1 Scale divisions, left scales.-The left scales shall be divided and marked as specified below. Scale markers shall be oriented to the left.

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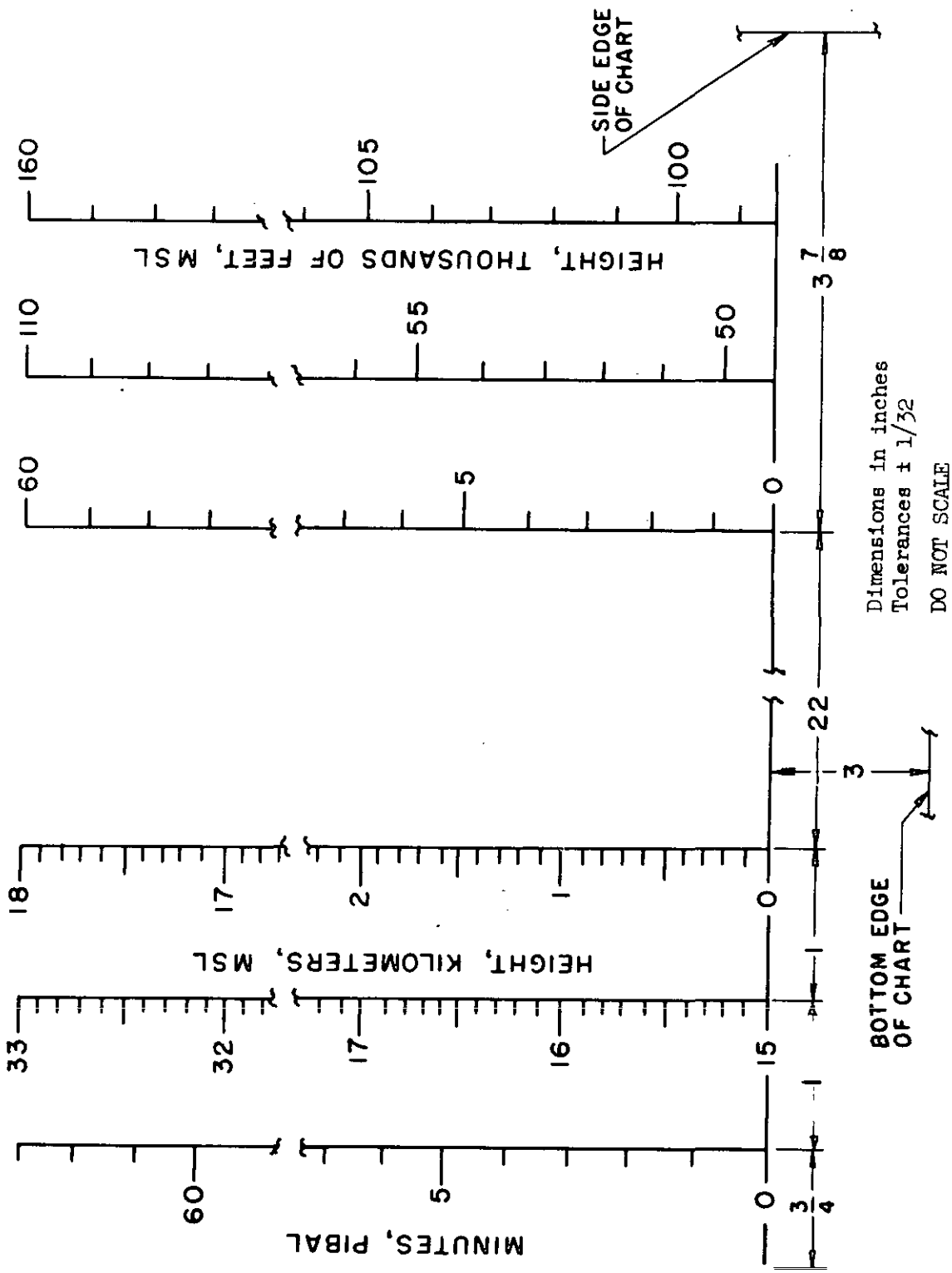


FIGURE 1 CHART SCALES

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a. *Number of scale divisions*

Inner and center scales	180
Outer scale	63

b. Distance between scale markers

Inner and center scales	0.133 inch
Outer scale	See table I

TABLE I

Minutes, Pibal, Scale

Division	Inches Above Ref. Line	Division	Inches Above Ref. Line	Division	Inches Above Ref. Line
1	0.467	22	8.393	43	16.247
2	0.893	23	8.753	44	16.640
3	1.307	24	9.113	45	17.033
4	1.713	25	9.473	46	17.433
5	2.113	26	9.833	47	17.833
6	2.507	27	10.193	48	18.233
7	2.893	28	10.553	49	18.633
8	3.273	29	10.913	50	19.033
9	3.653	30	11.273	51	19.427
10	4.027	31	11.640	52	19.813
11	4.400	32	12.007	53	20.193
12	4.773	33	12.380	54	20.567
13	5.140	34	12.753	55	20.940
14	5.507	35	13.133	56	21.313
15	5.873	36	13.513	57	21.687
16	6.233	37	13.893	58	22.060
17	6.593	38	14.280	59	22.433
18	6.953	39	14.673	60	22.807
19	7.313	40	15.067	61	23.180
20	7.673	41	15.460	62	23.560
21	8.033	42	15.853	63	23.933

c. Scale marker lengths

Each tenth marker	0.25 inch
Each fifth marker	0.20 inch
Other markers	0.10 inch

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d. Scale markers to be labeled

Inner and center scales	Each tenth marker
Outer scale	Each fifth marker

3.4.2.3.2 Scale divisions, right scales.-The right scales shall be divided and marked as specified below. Scale marker orientation shall be to the right.

a. Number of scale divisions

Inner scale	60
Center scale	60 (See note 1)
Outer scale	61 (See note 2)

b. Distance between scale markers

All scales	0.406 inch

c. Scale marker lengths

Each fifth marker	0.20 inch
Other markers	0.10 inch

d. Markers to be labeled

All scales	Each fifth marker

NOTES: 1. The center scale shall have its first marker (0.20 inch) at 0.320 inch above the base reference line and this marker shall be labeled 50.

2. The outer scale shall have a 0.10 inch marker at 0.234 inch above the base reference line. This marker shall not be labeled, but the next marker (0.20 inch) shall be labeled 100.

3.4.2.3.3 Scale identifiers.-The scales shall be identified as shown in figure 1. Labeling shall be equidistant from the top and bottom chart edges.

3.4.2.3.4 Labeling.-Labels shall be clear and sharp. Characters (letters and numerals) shall be approximately 7/64 inch high and shall have a height-to-width ratio of approximately 3 to 2. Spacing between characters shall be approximately one-third character width.

3.4.2.3.5 Lines.-Lines shall be clear, sharp, and unbroken except where otherwise specified. Lines and scale markers shall be approximately 0.020 inch wide.

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3.4.2.3.6 Scale division tolerances.-All scale markers shall be positioned within 0.005 inch of their true calculated position.

3.4.2.3.7 Scale marker length tolerances.-Scale marker length tolerances shall be plus or minus 0.010 inch.

3.4.2.3.8 Colors.-All labels, scale markers, and lines shall be dull black, unless otherwise specified.

3.4.2.3.9 Parallelism.-The parallelism of the six scales with respect to each other and the sides of the chart shall be sufficiently accurate to permit compliance with 3.4.2.5. The perpendicularity of the horizontal base line with respect to the six scales shall be equally accurate.

3.4.2.4 Plastic sheets.-The plastic sheets shall be made of vinyl chloride acetate in accordance with L-V-351. Each sheet shall be not less than 0.010 inch thick. The upper sheet shall be clear and transparent and free from cracks, seams, bubbles, raised spots, blemishes, and discoloration after the lamination process, so that the chart scales are clearly and distinctly visible. In addition, the top surface of the upper sheet shall be smooth, level, and finished in such a manner that sharp legible pencil and ink marks may be made on the plotting surface. It shall be possible to erase pencil lines completely, or until only a faint line remains, using a soft lead-pencil eraser. All marks shall be easily removable when the surface is washed with soap and water. The surface shall not be marred by repeated erasures, or repeated movement of the straightedge up and down the board.

3.4.2.5 Assembly of base chart to plotting board.-The laminated base chart shall be permanently secured to the plotting board with a suitable adhesive. The alignment of the chart on the board shall be such that, when the straightedge and frame are assembled to the board, the vertical reference line (3.4.3.4.1) shall coincide with the left inner scale (0-18KM) of the chart over the entire length of this scale, within 1/64 inch. In addition, the beveled plotting edge and the lower horizontal reference line (3.4.3.4.3) of the straightedge shall coincide within 1/64 inch, over their entire lengths, with the horizontal line at the base of the 6 chart scales when the straightedge is advanced to the appropriate positions. The 6 chart scales shall be perpendicular to the lower horizontal reference line (3.4.3.4.3) of the straightedge within one-tenth degree as the straightedge is moved up and down the board.

3.4.2.5.1 Bonding.-The bonding of the laminated chart to the baseboard shall be uniform and permanent over the entire

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d. Scale markers to be labeled

Inner and center scales	Each tenth marker
Outer scale	Each fifth marker

3.4.2.3.2 Scale divisions, right scales.-The right scales shall be divided and marked as specified below. Scale marker orientation shall be to the right.

a. Number of scale divisions

Inner scale	60
Center scale	60 (See note 1)
Outer scale	61 (See note 2)

b. Distance between scale markers

All scales	0.406 inch

c. Scale marker lengths

Each fifth marker	0.20 inch
Other markers	0.10 inch

d. Markers to be labeled

All scales	Each fifth marker

NOTES: 1. The center scale shall have its first marker (0.20 inch) at 0.320 inch above the base reference line and this marker shall be labeled 50.

2. The outer scale shall have a 0.10 inch marker at 0.234 inch above the base reference line. This marker shall not be labeled, but the next marker (0.20 inch) shall be labeled 100.

3.4.2.3.3 Scale identifiers.-The scales shall be identified as shown in figure 1. Labeling shall be equidistant from the top and bottom chart edges.

3.4.2.3.4 Labeling.-Labels shall be clear and sharp. Characters (letters and numerals) shall be approximately 7/64 inch high and shall have a height-to-width ratio of approximately 3 to 2. Spacing between characters shall be approximately one-third character width.

3.4.2.3.5 Lines.-Lines shall be clear, sharp, and unbroken except where otherwise specified. Lines and scale markers shall be approximately 0.020 inch wide.

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3.4.2.3.6 Scale division tolerances.-All scale markers shall be positioned within 0.005 inch of their true calculated position.

3.4.2.3.7 Scale marker length tolerances.-Scale marker length tolerances shall be plus or minus 0.010 inch.

3.4.2.3.8 Colors.-All labels, scale markers, and lines shall be dull black, unless otherwise specified.

3.4.2.3.9 Parallelism.-The parallelism of the six scales with respect to each other and the sides of the chart shall be sufficiently accurate to permit compliance with 3.4.2.5. The perpendicularity of the horizontal base line with respect to the six scales shall be equally accurate.

3.4.2.4 Plastic sheets.-The plastic sheets shall be made of vinyl chloride acetate in accordance with L-V-351. Each sheet shall be not less than 0.010 inch thick. The upper sheet shall be clear and transparent and free from cracks, seams, bubbles, raised spots, blemishes, and discoloration after the lamination process, so that the chart scales are clearly and distinctly visible. In addition, the top surface of the upper sheet shall be smooth, level, and finished in such a manner that sharp legible pencil and ink marks may be made on the plotting surface. It shall be possible to erase pencil lines completely, or until only a faint line remains, using a soft lead-pencil eraser. All marks shall be easily removable when the surface is washed with soap and water. The surface shall not be marred by repeated erasures, or repeated movement of the straightedge up and down the board.

3.4.2.5 Assembly of base chart to plotting board.-The laminated base chart shall be permanently secured to the plotting board with a suitable adhesive. The alignment of the chart on the board shall be such that, when the straightedge and frame are assembled to the board, the vertical reference line (3.4.3.4.1) shall coincide with the left inner scale (0-18KM) of the chart over the entire length of this scale, within 1/64 inch. In addition, the beveled plotting edge and the lower horizontal reference line (3.4.3.4.3) of the straightedge shall coincide within 1/64 inch, over their entire lengths, with the horizontal line at the base of the 6 chart scales when the straightedge is advanced to the appropriate positions. The 6 chart scales shall be perpendicular to the lower horizontal reference line (3.4.3.4.3) of the straightedge within one-tenth degree as the straightedge is moved up and down the board.

3.4.2.5.1 Bonding.-The bonding of the laminated chart to the baseboard shall be uniform and permanent over the entire

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surface area. The bonding process shall not cause separations or bubbles to occur in the laminated base chart.

3.4.2.5.2 Flatness.-There shall be no evidence of waves or ripples on the plotting surface, after the laminated chart is bonded to the baseboard, when the plotting surface is viewed at an angle of approximately 30 degrees to the line of sight of the observer. The total deviation from a flat surface shall not exceed one-sixteenth inch in width or length.

3.4.3 Straightedge

3.4.3.1 Material.-The straightedge shall be made of clear transparent vinyl chloride acetate, in accordance with L-V-351. It shall be approximately 3/16 inch thick and shall be 30 inches long plus or minus 1/32 inch. The straightedge width shall be as specified in 3.4.3.3.

3.4.3.2 Brackets.-The straightedge shall be equipped with two metal brackets of the adjustable-clamp type which shall permit it to be connected to the parallel attachment of the baseboard. Each bracket shall be attached to the straightedge by means of a screw positioned upwards through the plastic and a knurled nut. The outside diameter of the knurled portion of the nut shall be approximately 3/4 inch. The straightedge shall be easily detachable from the plotting board by removing the knurled nuts. The right hand bracket shall be spring-loaded to prevent side motion of the straightedge. The brackets shall be slotted to permit alignment of the straightedge with respect to the chart.

3.4.3.3 Straightedge width.-The straightedge shall have a width of 2 1/4 inches over the distance from its right edge to the vertical reference line (3.4.3.4.1). Over the distance from this line to the left edge of the straightedge, the width shall be 2 11/16 inches, except for the transitional 1/8 inch radius shown in figure 2.

3.4.3.4 Straightedge markings.-The straightedge shall be provided with three scales, upper, middle and lower. The straightedge shall also be provided with a vertical reference line, an upper horizontal reference line, and a lower horizontal reference line. These scales and lines shall be as shown in figure 2 and herein and shall be printed on the underside of the scale. They shall be protected from obliteration by a lamination approximately 0.010 inch thick. This lamination shall be bonded evenly to the scale over its entire underside, and there shall be no evidence of separation of the lamination from the scale.

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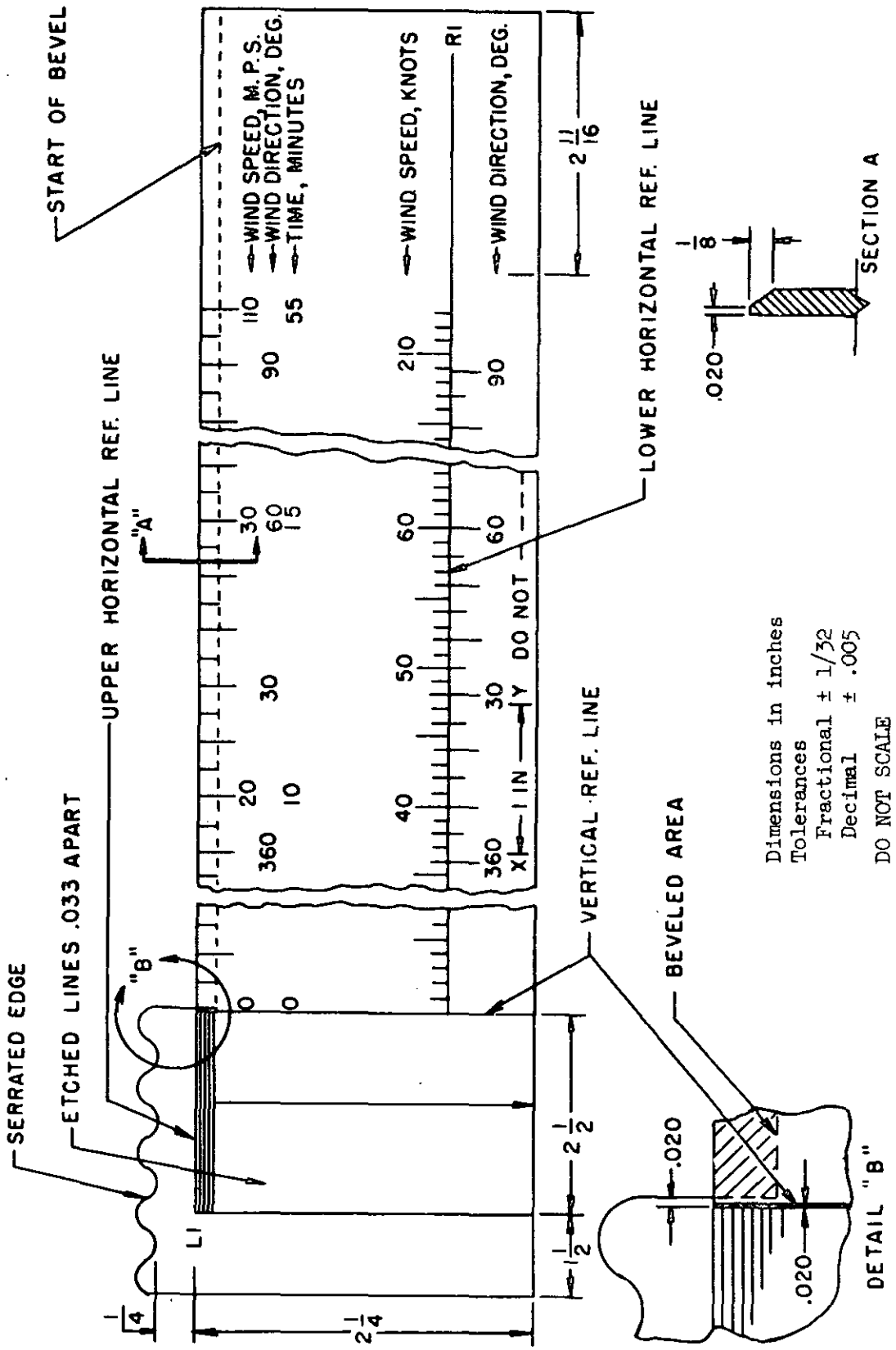


FIGURE 2 STRAIGHTEDGE

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3.4.3.4.1 Vertical reference line.-A vertical reference line shall be placed on the straightedge approximately 4 inches from the left side. This line shall coincide with the left inner scale on the chart (0-18KM) when the straightedge is assembled to the plotting board. The vertical reference line shall extend from the bottom of the straightedge to the top.

3.4.3.4.2 Upper horizontal reference line.-The upper horizontal reference line shall be an extension of the upper plotting edge and shall be engraved or etched from the vertical reference to a point approximately 2 1/2 inches to the left of the vertical reference line. The letters LI shall appear opposite the upper horizontal reference line approximately 1/2 inch to the left of its termination point.

3.4.3.4.3 Lower horizontal reference line.-The lower horizontal reference line shall be placed 0.6 inch plus or minus 0.015 inch above the bottom edge of the straightedge. It shall extend from the vertical reference line to a point approximately 1 1/2 inches from the right side of the straightedge. This line shall separate the middle and lower scales. The letters RI shall appear to the right of this line, 1/2 inch beyond its termination point.

3.4.3.4.4 Additional lines.-An additional set of horizontal lines shall be etched or engraved at the left side of the straightedge between the vertical reference line and a point 2 1/2 inches to the left. These lines shall be spaced at intervals of 0.033 inch between the upper horizontal reference line and the bottom of the straightedge. With the exception of the upper horizontal reference line, these lines shall not be inked or pigmented. A vertical line, parallel to the vertical reference line, shall inclose these lines.

3.4.3.4.5 Details of upper scale.-The upper scale shall coincide with the beveled top edge of the straightedge, extending from the vertical reference line for a distance of 22 inches to the right. The upper scale shall be divided and marked as follows:

a.	Number of scale divisions	110
b.	Distance between scale markers	0.20 inch
c.	Scale marker orientation	Down
d.	Scale marker lengths	
	Each tenth scale marker	0.25 inch
	Each even scale marker	0.20 inch
	Other scale markers	0.10 inch

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e. Scale marker to be labeled

Wind speed, M.P.S. line	See note 1
Wind direction, Deg. line	See note 2
Time, Minutes, line	See note 3

f. Line identification.-The scale divisions of the upper scale shall be capable of being interpreted in three ways. This shall be accomplished by marking three line identifiers at the right side of the scale. The line identifiers shall be arranged in columnar fashion, separated by approximately 1/16 inch, in the order indicated below. The left edge of the column shall be 2 11/16 inches plus or minus 1/32 inch from the right edge of straightedge. The lettering shall not be affected by the notches (3.4.3.5.3).

← WIND SPEED, M.P.S.
← WIND DIRECTION, DEG.
← TIME, MINUTES

NOTES:

1. The tenth scale marker to the right of the vertical reference line on the Wind speed, M.P.S. line shall be labeled with the value 10. Each successive tenth scale marker to the right shall be labeled with a value which is greater than the preceding value by an increment of 10. The last scale marker on this line shall be labeled 110.

2. Labeling for the Wind Direction, Deg. line including the identifier shall be in red. The figure 360 shall appear below the eighteenth scale marker to the right of the vertical reference line. The figure 30 shall appear below the twenty-fourth scale marker. Beneath each successive sixth scale marker, progressing to the right of the scale, a figure shall appear which is greater than the preceding figure by an increment of 30, until the value of 360 is reached. The next value shall be 30. The value of 90 shall be the last value to appear at the right, beneath the one hundred-eighth scale marker.

3. Each tenth scale marker on the time, minutes line shall be labeled with one-half the values used for the same scale marker on the wind speed, M.P.S. line.

4. The value 0 shall be placed at the vertical reference line for the wind speed, M.P.S. and time, minutes lines only.

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3.4.3.4.6 Details of middle scale.-The middle scale shall be placed along the upper edge of the lower horizontal reference line. The middle scale shall be divided and labeled as follows:

a.	Number of scale divisions	213
b.	Distance between scale markers	0.103 inch
c.	Scale marker orientation	Up
d.	Scale marker lengths	
	Each tenth scale marker	0.250 inch
	Each fifth scale marker	0.200 inch
	Other scale markers	0.100 inch

e. Scale marker labeling.-The tenth scale marker to the right of the vertical reference line on the Wind Speed, Knots scale shall be labeled with the value 10. Each successive tenth scale marker to the right shall be labeled with a value which is greater than the preceding value by an increment of 10. The highest value label appearing on this line shall be 210.

f. Scale identification.-The following scale identifier shall be placed to the right of and in line with the scale marker of the middle scale, and directly beneath the line identifiers for the upper scale:



3.4.3.4.7 Details of lower scale.-The lower scale shall be placed along the lower edge of the lower horizontal reference line. The labels including the identifier (see g.) shall be inked in red. This scale shall be divided and labeled as follows:

a.	Number of scale divisions	90
b.	Distance between scale markers	0.200 inch
c.	Location of left end of scale	3.600 inches to the right of vertical reference line
d.	Scale marker orientation	Down
e.	Scale marker lengths	
	1st, 3rd, 5th, etc.	0.200 inch
	2nd, 4th, 6th, etc.	0.100 inch

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f. Scale markers to be labeled	Scale marker labeling shall be identical with that specified for the Wind Direction, Deg line of the upper scale

g. Scale identification.-The following scale identifier shall be placed to the right of and in line with the scale markers of the lower scale, and directly beneath the line identifiers for the scales above.

←——— WIND DIRECTION, DEG.

3.4.3.4.8 Other straightedge marking details shall be in accordance with the following requirements:

	See Requirement
a. Labeling	3.4.2.3.4
b. Lines	3.4.2.3.5
c. Scale division tolerances	3.4.2.3.6
d. Scale marker length tolerances	3.4.2.3.7
e. Colors	3.4.2.3.8

f. Scale and line identifier arrows.-Line and scale identifier arrows shall be 3/16 inch long plus or minus 1/32 inch.

3.4.3.5 *Configuration of straightedge*

3.4.3.5.1 Flatness.-The straightedge shall not be bowed or warped. It shall be flat within 1/64 inch when laid on a surface plate or similar flatness testing device.

3.4.3.5.2 Serrations and bevel.-The upper edge of the straightedge shall be serrated and beveled as shown in figure 2.

3.4.3.5.3 Notches.-The left and right sides of the bottom surface of the straightedge shall be notched to accommodate the frame. These notches shall be of the minimum width and height necessary to clear the frame and meet the design requirements of this specification. They shall not interfere with the operation of the parallel mechanism, and shall not introduce unnecessary side motion in the straightedge.

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3.4.3.5.4 Bracket attachment holes.-The straightedge shall be provided with a hole on each end for attachment of the brackets (3.4.3.2).

3.4.3.6 Inspection gauge.-An inspection gauge shall be printed or engraved on the straightedge. It shall be centered between the left and right sides of the straightedge and between the bottom of the straightedge and the labels for the lower scale. The gauge shall consist of two parallel vertical lines, directly opposite each other, 1/8 inch high, plus or minus 1/32 inch, and 1 inch, plus or minus 1/64 inch apart. These lines shall be labeled "X" and "Y" and the 1 inch dimension shall appear between them as shown in figure 2. The following notation shall appear to the right of the gauge: DO NOT USE STRAIGHTEDGE IF X-Y DIMENSION IS NOT INCH PLUS OR MINUS 1/64. The inspection gauge and notation shall be colored red. Labeling and lines shall be as specified in 3.4.2.3.4 and 3.4.2.3.5.

3.4.4 Frame.-A suitable metal frame shall be provided for the purpose of protecting the edges of the plotting board and base chart. This frame shall be constructed of 1/16 inch thick angle material, with a maximum side width of 1/2 inch. The metal used shall meet the requirements of 3.3.3 and, in addition, shall be chosen so that the surface of the frame shall not be marred by continuous movement of the straightedge over the frame. The frame shall be furnished in four pieces 30 1/8 inches long plus or minus 1/32 inch. The ends of the frame pieces shall be cut to permit mitered corners. The four pieces shall be secured to the plotting board by means of flat head screws. The joints of the frame shall be smooth after assembly to the board, and free from burrs, sharp or overlapping edges or gaps in excess of 1/16 inch. The countersunk holes in the frame shall be free of burrs and the screws shall be sufficiently recessed to prevent marring of the straightedge in use.

3.5 Protection for cords.-The cords on the bottom of the board shall be protected from damage by metal shields, approximately 1/4 inch wide, or cross-section similar to the Greek capital letter omega (Ω), and attached to the bottom of the board by screws. The shields shall cover the segments of the cords' paths which are diagonal. No more than 2 inches of cord, starting at each pulley and going toward the center of the board, shall be uncovered.

NOTE: At the center of the board, no more than 1 inch of cord, starting at the point of crossing and going toward each corner of the board, shall be uncovered. The segments which are parallel to the edges of the board shall not be covered.

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3.6 Service conditions.-The plotting board shall retain the specified degree of flatness, permanency of bonding, freedom from warpage and corrosion, etc., when subjected to temperatures varying from 0 degrees Centigrade (C) to 52 degrees C and relative humidities varying from 0 to 100 percent. The plotting board shall withstand storage under temperature conditions ranging from minus 40 degrees C to plus 54 degrees C and relative humidity varying from 0 to 100 percent.

3.7 Interchangeability.-All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-D-70327 shall govern changes in the manufacturer's part numbers.

3.7.1 Interchangeability assurance.-The Plotting Board, Winds Aloft ML-514/TM fabricated under this specification shall be interchangeable part by part with the equipment loaned by the Government to the contractor (3.9). The loan of the Plotting Board, Winds Aloft ML-514/TM is made solely for the purpose of interchangeability assurance and such equipment shall serve as a reference standard for interchangeability as defined in MIL-STD-447.

3.8 Identification of product.-Nomenclature assignments and approval of nameplate identification for the product shall be obtained in accordance with MIL-N-7513. The equipment marking shall be printed on the paper chart, in the upper left-hand corner, within a block approximately 2 inches high and 4 inches wide.

3.9 Government-loaned property.-When the contract or purchase order so provides, the Government will loan the following to the contractor upon his request for the purpose of interchangeability assurance (3.7.1):

Item No.	Stock No.	Item Description	Quantity
1	6660-557-0766	Plotting Board, Winds Aloft ML-514/TM	1 each

3.10 Workmanship.-The plotting board, including all parts and accessories, shall be constructed in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes and defects, accuracy of dimensions, flatness, straightedges, markings, freedom of parts from burrs and sharp edges, thoroughness of bonding, clearness and sharpness of lines and labels, tightness of screws, and quality of the lamination process.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Classification of tests.-The inspection and testing of the Plotting Board, Winds Aloft ML-514/TM shall be classified as follows:

a. Acceptance tests (4.3)

b. Preproduction tests (4.6)

4.3 Acceptance tests.-Acceptance tests shall consist of:

a. Individual tests (4.3.1)

4.3.1 Individual tests.-Each Winds Aloft Plotting Board ML-514/TM, shall be subjected to the following tests as described under 4.5, Test methods, of this specification:

a. Examination of product (4.5.1)

b. Functional tests (4.5.2)

4.4 Test conditions

4.4.1 Atmospheric conditions.-Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of 28 to 32 inches of mercury and at a temperature of 25 degrees C, plus or minus 10 degrees C, and a relative humidity of 80 percent or less. Where tests are made with atmospheric pressure or temperature substantially different from the above values, proper allowances shall be made for the changes in instrument reading.

4.5 Test methods

4.5.1 Examination of product.-Each Winds Aloft Plotting Board ML-514/TM shall be inspected to determine compliance with this specification with respect to material, dimensions, marking, workmanship, and conformance to figures.

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4.5.2 Functional tests.-Each plotting board shall be inspected to determine conformance to the following functional requirements of this specification:

a.	Pencil and ink marks	(3.4.2.4)
b.	Chart alignment	(3.4.2.5)
c.	Brackets	(3.4.3.2)
d.	Notches	(3.4.3.5.3)

4.5.3 Performance tests.-The plotting boards shall be tested for compliance with the following:

a.	Chart scales	(3.4.2.3)
b.	Straightedge markings	(3.4.3.4)
c.	Resistance of frame to marring	(3.4.4)
d.	Resistance of plotting surface to marring	(3.4.2.4)
e.	Service conditions	(3.6)
f.	Interchangeability	(3.7)

4.5.3.1 Details of chart and straightedge scales.-Complete compliance with all detailed scale requirements (3.4.2.3, 3.4.3.4) shall be verified by measurements with appropriate gauges and measuring tools.

4.5.3.1.1 Master dies and plates.-If the contractor uses master dies, plates, or similar production tools for uniform reproduction of the scales on each plotting board to be submitted under contract, verification of the accuracy of the scale impressions on the preproduction samples may be considered adequate for determining compliance with the scale accuracy requirements. Inspection of scale dimensions in the acceptance tests (4.3) may be limited to inspection for clearness and sharpness of printed lines and labels, and proper location of printing on the chart and the straightedge. If master dies, plates, etc., are not used for the production item, inspection templates shall be used in the acceptance tests.

4.5.3.2 Resistance of frame and plotting surface to marring.-A suitable test shall be performed to insure compliance with the requirements of 3.4.2.4 and 3.4.4 with respect to the effect of

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4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Classification of tests.-The inspection and testing of the Plotting Board, Winds Aloft ML-514/TM shall be classified as follows:

a. Acceptance tests (4.3)

b. Preproduction tests (4.6)

4.3 Acceptance tests.-Acceptance tests shall consist of:

a. Individual tests (4.3.1)

4.3.1 Individual tests.-Each Winds Aloft Plotting Board ML-514/TM, shall be subjected to the following tests as described under 4.5, Test methods, of this specification:

a. Examination of product (4.5.1)

b. Functional tests (4.5.2)

4.4 Test conditions

4.4.1 Atmospheric conditions.-Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of 28 to 32 inches of mercury and at a temperature of 25 degrees C, plus or minus 10 degrees C, and a relative humidity of 80 percent or less. Where tests are made with atmospheric pressure or temperature substantially different from the above values, proper allowances shall be made for the changes in instrument reading.

4.5 Test methods

4.5.1 Examination of product.-Each Winds Aloft Plotting Board ML-514/TM shall be inspected to determine compliance with this specification with respect to material, dimensions, marking, workmanship, and conformance to figures.

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4.5.2 Functional tests.-Each plotting board shall be inspected to determine conformance to the following functional requirements of this specification:

a.	Pencil and ink marks	(3.4.2.4)
b.	Chart alignment	(3.4.2.5)
c.	Brackets	(3.4.3.2)
d.	Notches	(3.4.3.5.3)

4.5.3 Performance tests.-The plotting boards shall be tested for compliance with the following:

a.	Chart scales	(3.4.2.3)
b.	Straightedge markings	(3.4.3.4)
c.	Resistance of frame to marring	(3.4.4)
d.	Resistance of plotting surface to marring	(3.4.2.4)
e.	Service conditions	(3.6)
f.	Interchangeability	(3.7)

4.5.3.1 Details of chart and straightedge scales.-Complete compliance with all detailed scale requirements (3.4.2.3, 3.4.3.4) shall be verified by measurements with appropriate gauges and measuring tools.

4.5.3.1.1 Master dies and plates.-If the contractor uses master dies, plates, or similar production tools for uniform reproduction of the scales on each plotting board to be submitted under contract, verification of the accuracy of the scale impressions on the preproduction samples may be considered adequate for determining compliance with the scale accuracy requirements. Inspection of scale dimensions in the acceptance tests (4.3) may be limited to inspection for clearness and sharpness of printed lines and labels, and proper location of printing on the chart and the straightedge. If master dies, plates, etc., are not used for the production item, inspection templates shall be used in the acceptance tests.

4.5.3.2 Resistance of frame and plotting surface to marring.-A suitable test shall be performed to insure compliance with the requirements of 3.4.2.4 and 3.4.4 with respect to the effect of

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repeated straightedge movement. This test shall employ a mechanical means to move the straightedge up and down the length of the plotting board continuously at least 500 times in each direction. The straightedge shall be preloaded to simulate operating hand pressures. There shall be no evidence of grooves, scratches, or other deterioration of the frame, straightedge, or plotting board surfaces.

4.5.3.3 Service conditions.-The plotting board shall be tested for compliance with the requirements of 3.6 as follows:

4.5.3.3.1 Operating conditions.-The plotting board shall be maintained at a temperature of 0 degrees C for a minimum of 2 hours. Immediately after removal from the 0 degrees C environment, the acceptance tests shall be repeated and there shall be no evidence of inability to meet these tests. The board shall then be maintained at a temperature of 52 degrees C for a period of 2 hours. The acceptance tests shall be repeated immediately after removal of the board from the 52 degrees C environment. The plotting board shall meet the requirements of the acceptance tests.

4.5.3.3.2 Storage conditions.-The following tests shall be conducted. Successful completion of the humidity test shall constitute compliance with high temperature storage requirements.

a. Humidity.-In accordance with procedure I of MIL-E-5272, except that the maximum temperature shall be 54 degrees C instead of the specified 71 degrees C and the number of cycles shall be reduced to two.

b. Low temperature.-The plotting board shall be maintained at minus 40 degrees C for a minimum of 6 hours. Upon return to temperatures within the operating range (0 degrees C to 52 degrees C) the plotting board shall meet all the requirements of the acceptance tests.

4.6 Preproduction testing

4.6.1 Preproduction test samples.-The preproduction test samples shall consist of two models representative of the production equipment. They shall be tested at a laboratory designated by the procuring activity or, when so stated in the contract, at the contractor's plant under the supervision of the procuring activity.

4.6.2 Preproduction tests.-Preproduction tests shall consist of all tests described under 4.5, Test methods.

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5. PREPARATION FOR DELIVERY

5.1 Levels of preservation and packaging

5.1.1 Level A.-The Winds Aloft Plotting Board ML-514/TM shall be preserved and packaged in accordance with method IA of MIL-P-116.

5.1.2 Level C.-The Winds Aloft Plotting Board ML-514/TM shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2 Levels of packing

5.2.1 Level A.-The Winds Aloft Plotting Board ML-514/TM preserved and packaged to meet 5.1.1 shall be packed in exterior shipping containers conforming to PPP-B-591, PPP-B-621, PPP-B-636, MIL-B-138, PPP-B-601, or MIL-B-10377. As far as practical, exterior containers shall be of uniform shape and size, be of minimum cube and tare consistent with the protection required, and contain identical quantities. The gross weight of each pack shall be limited to approximately 200 pounds. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. Containers shall be provided with a case liner conforming to MIL-L-10547 and shall be sealed in accordance with the appendix thereto. The case liner will not be required when the unit, intermediate, or exterior container conforms to PPP-B-636 and is sealed at all joints and seams, including manufacturer's joint, with tape conforming to PPP-T-60.

5.2.2 Level B.-Winds Aloft Plotting Board ML-514/TM preserved and packaged to meet 5.1.1 shall be packed in domestic type exterior containers meeting PPP-B-591, PPP-B-601, PPP-B-621, PPP-B-636, PPP-B-585, or MIL-B-10377. Exterior containers shall be of minimum cube and tare consistent with the protection required. As far as practical, exterior containers shall be of uniform shape and size and contain identical quantities. The gross weight of each pack shall be limited to approximately 500 pounds. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. When fiberboard containers are used, the fiberboard shall meet the special requirements table of PPP-B-636 as applicable.

5.2.3 Level C.-Packages which require over-packing for acceptance by the carrier shall be packed in exterior type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers

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shall meet Consolidated Freight Classification Rules or regulations of other common carriers as applicable to the mode of transportation.

5.3 Physical protection.-Cushioning, blocking, bracing, and bolting as required shall be in accordance with JAN-P-100 except that for domestic shipments, waterproofing requirements for cushioning materials and containers shall be waived. Drop tests of JAN-P-100 shall be waived when preservation, packaging, and packing of the item is for immediate use or when drop tests of MIL-P-116 are applicable.

5.4 Marking.-Interior and exterior containers shall be marked in accordance with MIL-STD-129. The shipment marking nomenclature shall be:

Plotting Board, Winds Aloft ML-514/TM.

6. NOTES

6.1 Use.-The Winds Aloft Plotting Board ML-514/TM is meteorological equipment designed for the following functions:

a. Plotting a time-height curve as a step in rawin data reduction.

b. Plotting wind directions and wind speed curves, as a function of height, from rawin or pibal observations to permit rapid determination of standard level wind data for transmission purposes. The plotting board incorporates a simple method of determining the height of any station above sea level.

6.2 Ordering data.-Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Point of testing.

c. Whether the item in 3.9 shall be Government-loaned property.

d. Selection of applicable levels of preservation and packaging, and packing.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have

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